

Ch 7: How do Organisms Reproduce?

Part 1 of 1 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION A - INTRODUCTION TO REPRODUCTION AND ITS IMPORTANCE

1. Meaning of reproduction

- Reproduction is the biological process by which new organisms, called offspring, are produced from their parents.

2. Why reproduction is necessary

- It ensures the production of new individuals.
- It replaces individuals that die naturally due to ageing, disease, predation or accidents.

3. Continuity of life and species survival

- Reproduction ensures continuity of life on Earth by maintaining the population of a species.
- It helps in the survival of species across generations.

4. Difference between growth and reproduction

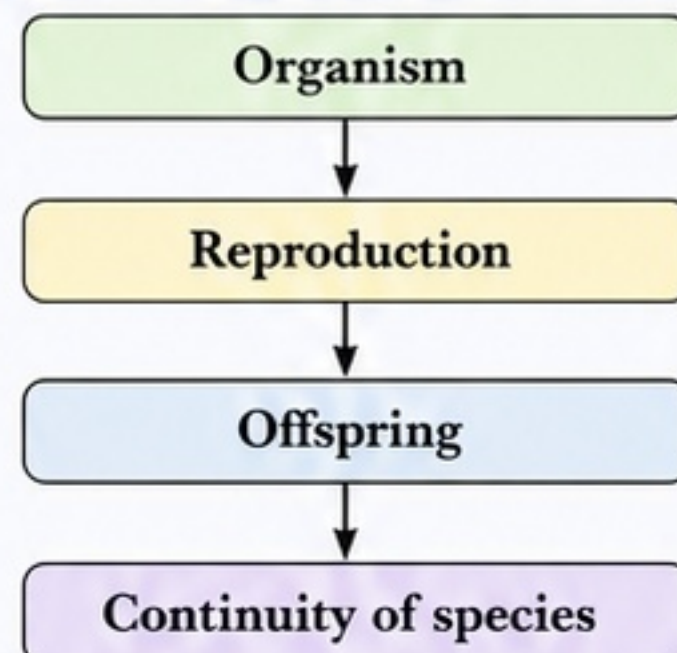
- Growth is an increase in size and body mass of an organism.
- Reproduction is the production of new individuals.
- Growth occurs in an individual; reproduction ensures new individuals.

5. Main types of reproduction

- Organisms reproduce in two broad ways:
 - Asexual reproduction** – involves a single parent; no fusion of gametes.
 - Sexual reproduction** – involves two parents (or two gametes) and fusion of gametes.

6. Broad comparison between asexual and sexual reproduction

FLOW DIAGRAM: CONTINUITY OF LIFE



Through reproduction, organisms produce offspring which grow, mature and reproduce, ensuring the continuity of their species.

8. BASIC TERMS

- Organism** – A living being (plant, animal, microbe, etc.).
- Offspring** – The young one produced by parents.
- Gamete** – A reproductive cell (sperm in males, ovum in females).
- Fertilisation** – The fusion of male and female gametes to form a zygote.
- Trait** – A specific characteristic of an organism, e.g., height, colour of flowers.
- Variation** – Differences in traits among individuals of the same species.

Basis	Asexual Reproduction	Sexual Reproduction
Number of parents	Single parent	Usually two parents (male and female)
Gametes involved	No gametes; no fusion	Gametes are formed and fused
Fertilisation	Absent	Present
Offspring	Genetically identical to parent (clones)	Genetically different from parents and from each other
Rate of process	Usually fast	Usually slow
Variation	No or very little variation	High variation
Adaptability	Low adaptability to changing environment	High adaptability
Examples	Binary fission (Amoeba), budding (Yeast), vegetative propagation (potato), spore formation (Mould)	Human beings, animals, most flowering plants

9. EVERYDAY EXAMPLES



In Plants

- A new plant grows from a cutting of rose, money plant or sugarcane (vegetative propagation).
- New plants also grow from spores in ferns.



In Animals

- Dogs give birth to puppies.
- Hens lay eggs; chicks hatch from eggs.
- Many animals, including humans, show parental care for their offspring.



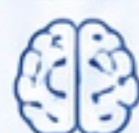
KEY IDEA

Reproduction is essential for the continuity of life. Asexual reproduction produces identical offspring, while sexual reproduction produces varied offspring due to the fusion of gametes.



EXAM TIPS

- Always start answers by defining reproduction.
- Write clear differences between growth and reproduction.
- Use diagrams and flowcharts wherever possible.
- In comparison tables, write crisp and point-wise lines.



QUICK RECALL

- Reproduction produces offspring.
- Ensures continuity of species.
- Two types: Asexual and Sexual.
- Asexual = no gametes, clones.
- Sexual = fusion of gametes, variation.
- More variation → better survival.

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SECTION B - ASEXUAL REPRODUCTION

1. Definition

- Asexual reproduction is a mode of reproduction in which a new individual is produced from a single parent **without the fusion of gametes**.

2. Key features

- Involves only one parent.
- No formation or fusion of gametes.
- Offspring are **genetically identical** (clones) to the parent.
- Usually occurs through body cells by simple cell division or specialized structures.

3. Advantages

- Requires only one parent; no mate is needed.
- Rapid and energy-efficient.
- Produces a large number of offspring in a short time.
- Helpful for colonization of new areas.

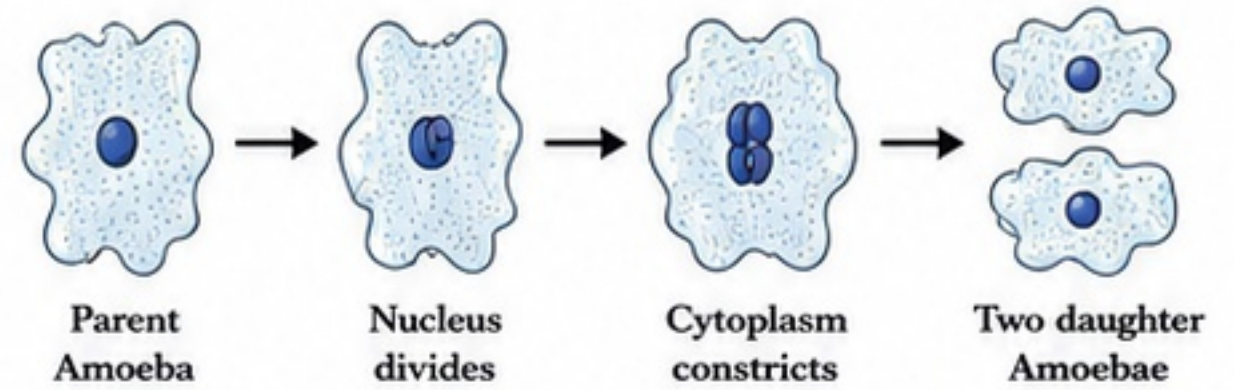
4. Limitations

- Little or no variation; all offspring are identical.
- Poor adaptability; cannot cope with environmental changes.
- Harmful traits may persist in the population.
- Overcrowding may occur due to rapid increase.

5. Methods of asexual reproduction

Method	Organism(s)	Key Point
Binary fission	Amoeba, <i>Leishmania</i>	Parent cell divides into two equal daughter cells.
Multiple fission	<i>Plasmodium</i>	Nucleus divides many times, followed by cytoplasmic division to form many offspring.
Budding	Yeast, <i>Hydra</i>	A small outgrowth (bud) develops on parent and forms a new individual.
Fragmentation	<i>Spirogyra</i>	Parent body breaks into pieces; each fragment grows into a new individual.
Regeneration	<i>Planaria</i>	A cut part regenerates the missing parts and forms a new organism.
Spore formation	<i>Rhizopus</i>	Spores formed in sporangia develop into new individuals.
Vegetative propagation	Plants (potato, sugarcane, rose, etc.)	New plants arise from vegetative parts such as root, stem or leaf.

DIAGRAM 1: BINARY FISSION IN AMOEBA

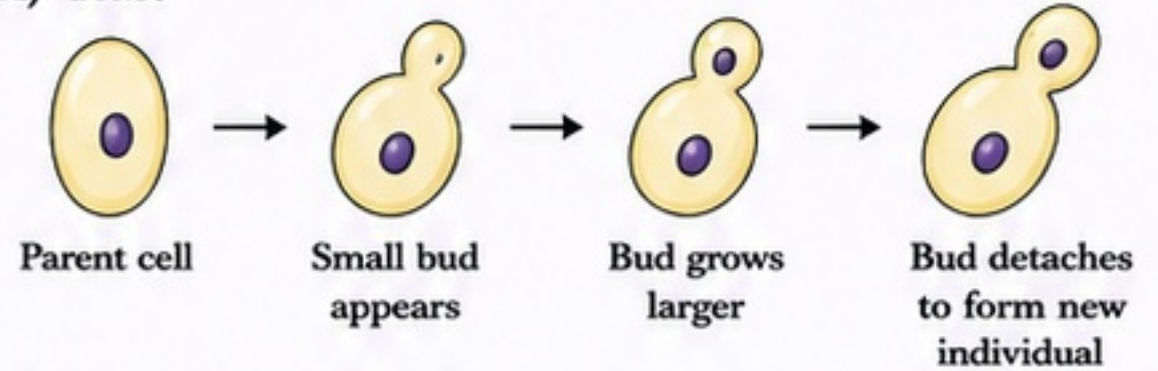


Parent divides into two equal daughter individuals.

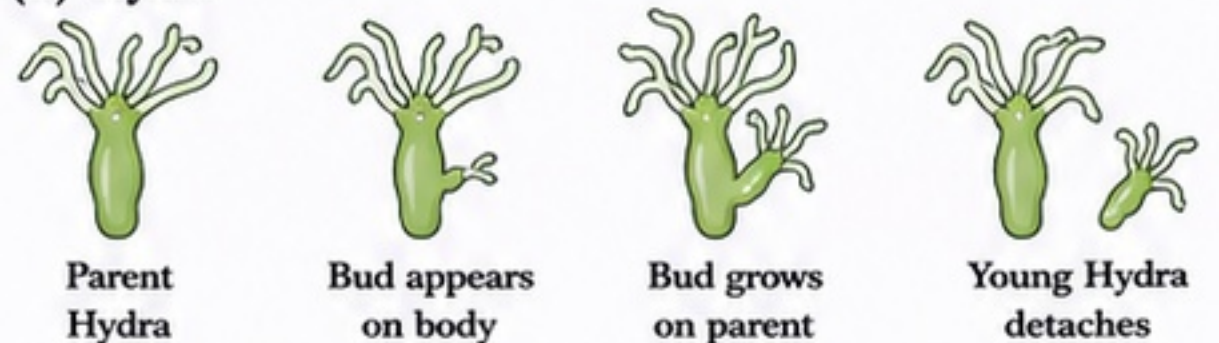
Example: *Amoeba*

DIAGRAM 2: BUDDING IN YEAST / HYDRA

(A) Yeast



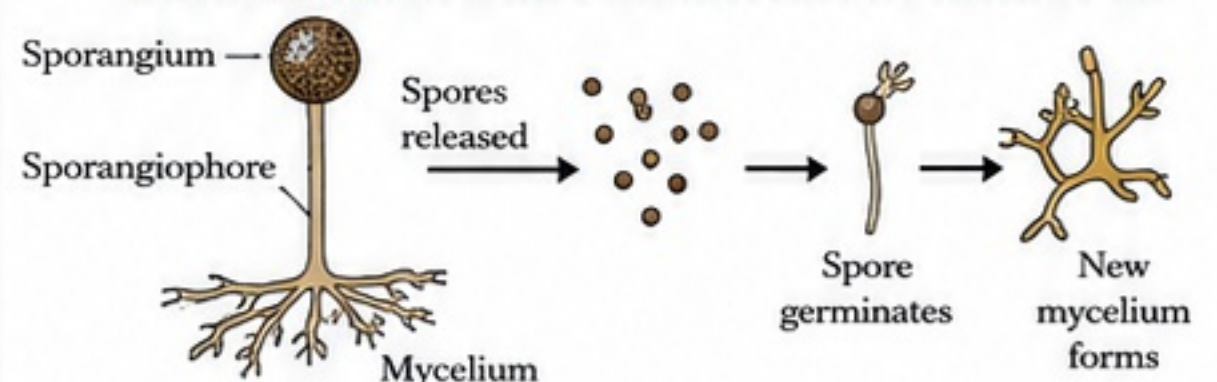
(B) Hydra



A small outgrowth (bud) develops on the parent, grows and detaches (or may remain attached).

Examples: *Yeast*, *Hydra*

DIAGRAM 3: SPORE FORMATION IN RHIZOPUS



Spores produced in sporangium are released, germinate under favourable conditions and develop into new individuals.

Example: *Rhizopus*



Note on Vegetative Propagation: In plants, new individuals arise from vegetative parts (root, stem or leaf) without gamete fusion. It may be natural (e.g., potato-tuber, ginger-rhizome, *Bryophyllum*-leaf buds) or artificial (e.g., cutting, layering, grafting). It is widely used in agriculture and horticulture for maintaining desirable varieties.

KEY IDEA

- Asexual reproduction involves one parent and produces genetically identical offspring (clones).
- It is quick and efficient but produces no variation.
- Occurs through different methods such as binary fission, budding, spore formation, regeneration and vegetative propagation.



EXAM TIPS

- Diagrams: *Amoeba* (binary fission), *Yeast/Hydra* (budding), *Rhizopus* (spore formation) are high probability.
- Write definition with key points: one parent, no gamete fusion, clones.
- Compare methods in a table for better marks.



QUICK RECALL

- One parent only.
- No fusion of gametes.
- Offspring are clones.
- Methods: Binary fission, Multiple fission, Budding, Fragmentation, Regeneration, Spore formation, Vegetative propagation.

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SECTION C - VEGETATIVE PROPAGATION IN PLANTS

1. What is vegetative propagation?

- Vegetative propagation is a type of asexual reproduction in plants in which new plants are produced from vegetative parts such as stem, root or leaf.
- The new plants are genetically identical to the parent plant (clones). Hence, it is an asexual method.

2. Natural vegetative propagation

A. By stem

- New plants arise from stems or their modifications.
- Examples:
 - Potato** – ‘eyes’ (buds) on the tuber sprout into new plants.
 - Ginger** – underground stem (rhizome) produces new shoots.
 - Onion** – fleshy underground buds (bulb) form new onion plants.
 - Sugarcane** – buds on the stem (nodes) grow into new plants.
 - Rose (sucker)** – shoots arise from the base/rootstock.

B. By root

- Adventitious buds arise from roots and form new plants.
- Example: Sweet potato** – swollen root gives rise to new plants.

C. By leaf

- Leaves or leaf parts develop adventitious buds along their margins or base.
- Example: Bryophyllum** – small plantlets form on the notches of the leaf; when they fall on soil, they grow into new plants.

3. Artificial vegetative propagation (done by humans)

A. Cutting

- A piece of stem or root with a bud is cut and planted in soil.
- Example: Rose, Bougainvillea, Hibiscus, Sugarcane.**

B. Layering

- A branch is bent to the ground, a part of it is covered with soil. Roots develop there. The rooted part is separated and planted.
- Example: Jasmine, Litchi, Guava.**

C. Grafting

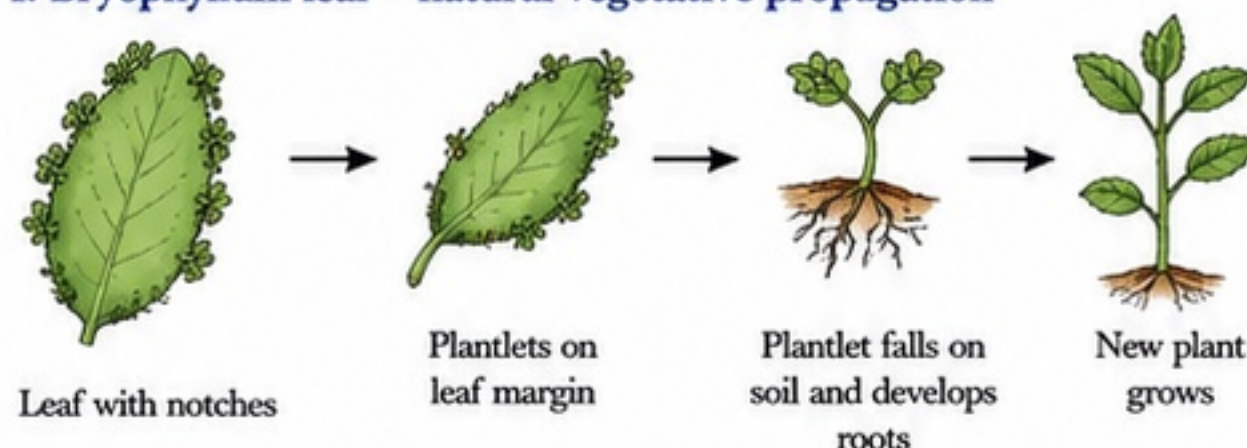
- A stem piece (scion) from a desired plant is joined to the rootstock of another plant so that they grow as one.
- Example: Mango, Rose, Citrus (orange, lemon).**

D. Tissue culture / Micropropagation

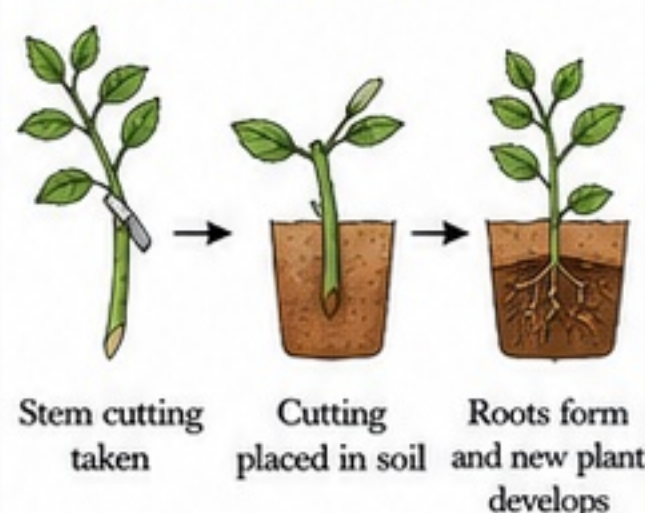
- Small pieces of plant tissue (cells, meristems) are grown in sterile nutrient medium in a laboratory.
- Many identical plants are produced in a short time.
- Example: Banana, Orchid, Sugarcane, Potato.**

DIAGRAMS

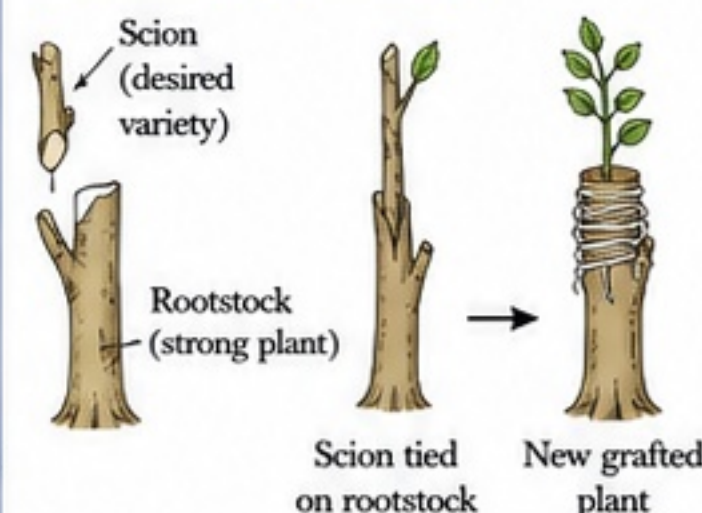
1. Bryophyllum leaf – natural vegetative propagation



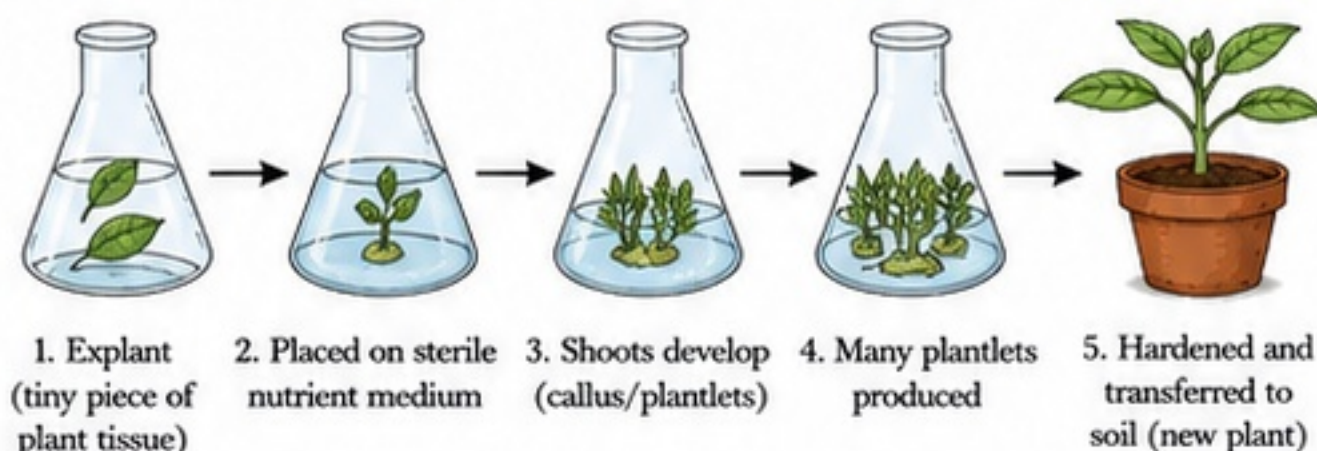
2. Cutting (e.g., Rose)



3. Grafting (e.g., Mango)



4. Tissue culture – micropropagation (steps)



Method	Plant Example	Plant Part Used	How It Works (in brief)	Result
Natural – By Stem	Potato	Tuber (stem ‘eyes’)	Eyes sprout and grow	New potato plant
Natural – By Stem	Ginger	Rhizome (underground stem)	Buds on rhizome produce shoots	New ginger plant
Natural – By Stem	Onion	Bulb (fleshy stem base)	Bulb scales form new bulbs	New onion plants
Natural – By Root	Sweet Potato	Swollen root	Adventitious buds form shoots	New sweet potato plant
Natural – By Leaf	Bryophyllum	Leaf	Plantlets form on leaf margins	New plantlets
Artificial – Cutting	Rose	Stem cutting	Roots develop on cutting	New rose plant
Artificial – Layering	Jasmine	Bent branch	Roots form on covered part	New jasmine plant
Artificial – Grafting	Mango	Scion + Rootstock	Both parts unite and grow	New grafted mango
Artificial – Tissue Culture	Banana	Plant tissue (meristem)	Grown in sterile medium	Many identical plantlets

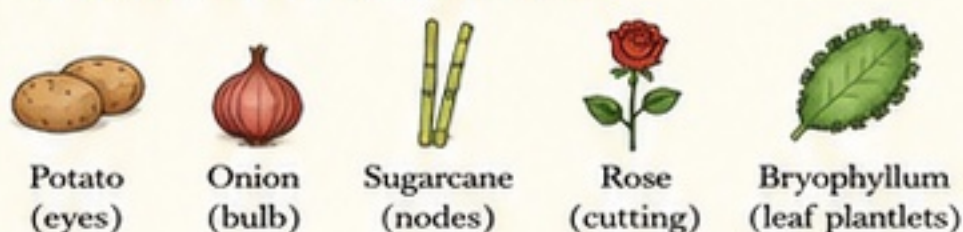
4. ADVANTAGES

- New plants are produced rapidly.
- Desirable characters of the parent are preserved.
- Plants bear flowers/fruit earlier than seed-grown plants.
- Useful for seedless plants (e.g., banana, seedless grapes).
- Helps in propagation of plants that do not produce seeds.

5. DISADVANTAGES

- No genetic variation – less ability to adapt to new conditions.
- Diseases of the parent can spread to all new plants.
- Plants may be weaker if the rootstock is poor.
- Requires skill and care in artificial methods.

6. EVERYDAY EXAMPLES



KEY IDEA

Vegetative propagation is an asexual method of reproduction in plants that uses vegetative parts (stem, root, leaf) to produce genetically identical new individuals quickly and efficiently.

EXAM TIPS

- Differentiate natural vs artificial methods.
- Give suitable examples for each method.
- Draw labelled diagrams (Bryophyllum plantlets, grafting, tissue culture).
- Write advantages and disadvantages clearly.



9. QUICK RECALL

- Vegetative parts: stem, root, leaf.

- Vegetative parts: stem, root, leaf.
- Natural methods: stem (potato, ginger, onion, sugarcane); root (sweet potato); leaf (Bryophyllum).
- Artificial methods: cutting, layering, grafting, tissue culture.
- Outcome: identical new plants; no seeds or gametes involved.

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SECTION D - SEXUAL REPRODUCTION IN FLOWERING PLANTS

1. Meaning of sexual reproduction in plants

- It involves two parents (male and female) and fusion of gametes.
- Produces genetic variation and forms seeds and fruits.

2. Structure of a flower

- A typical flower has four main whorls—sepals, petals, stamens and pistil (carpel).
- Floral parts may vary in number and form in different plants.

3. Male and female gametes

- Male gamete** – Pollen grain (produced in the anther).
- Female gamete** – Egg cell (present inside the ovule in the ovary).

4. Pollination

- Transfer of pollen grains from the anther to the stigma.
- It is the first step towards fertilisation.

5. Types of pollination

- Self-pollination** – Pollen transfer occurs within the same flower or between flowers of the same plant.
- Cross-pollination** – Pollen transfer occurs between flowers of different plants of the same species.

6. Agents/means of pollination

- Wind (e.g., grasses, maize)
- Water (e.g., *Vallisneria*)
- Animals (insects, birds, bats, etc.)

7. Fertilisation

- Fusion of male gamete (from pollen) with female gamete (egg cell) in the ovule.

8. Zygote and embryo

- The fused cell is called **zygote**.
- Zygote divides and develops into **embryo**.

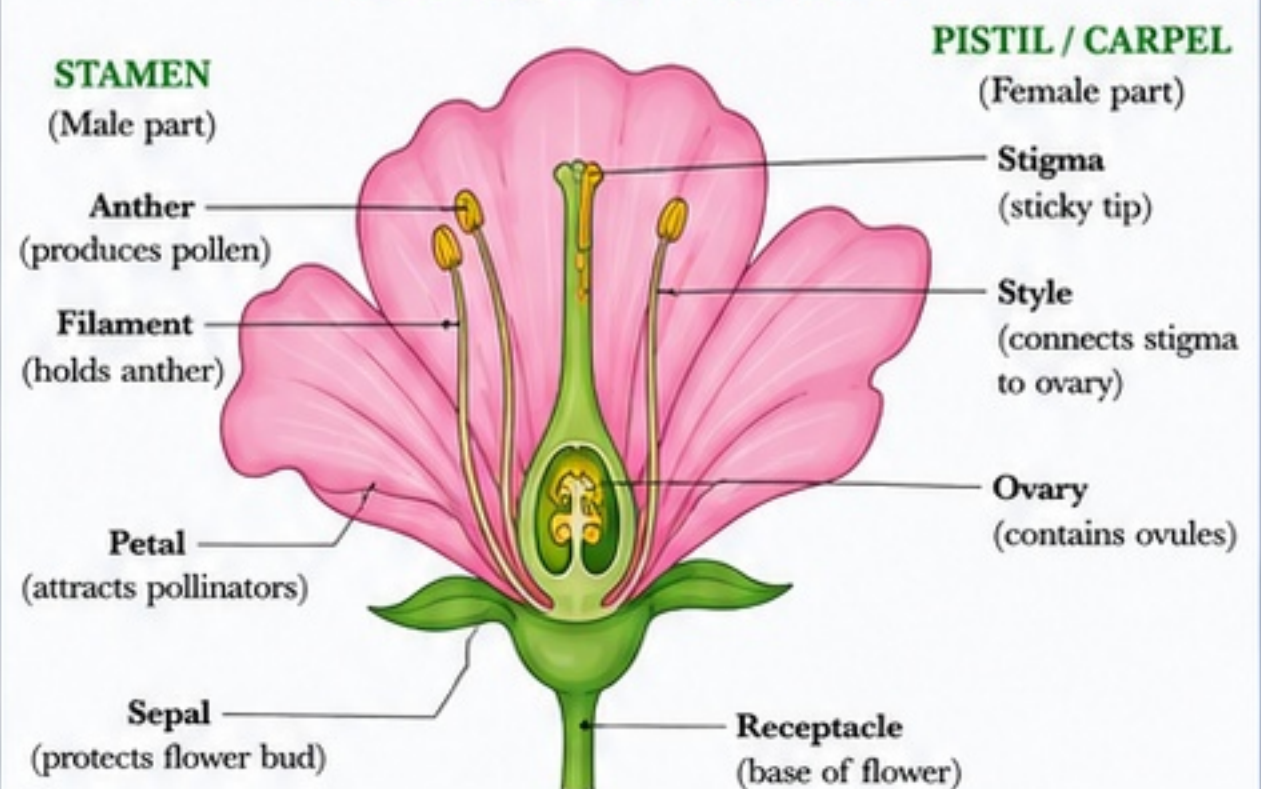
9. From ovule to seed

- After fertilisation, the ovule develops into a seed.
- Seed contains an embryo, food reserve and seed coat.

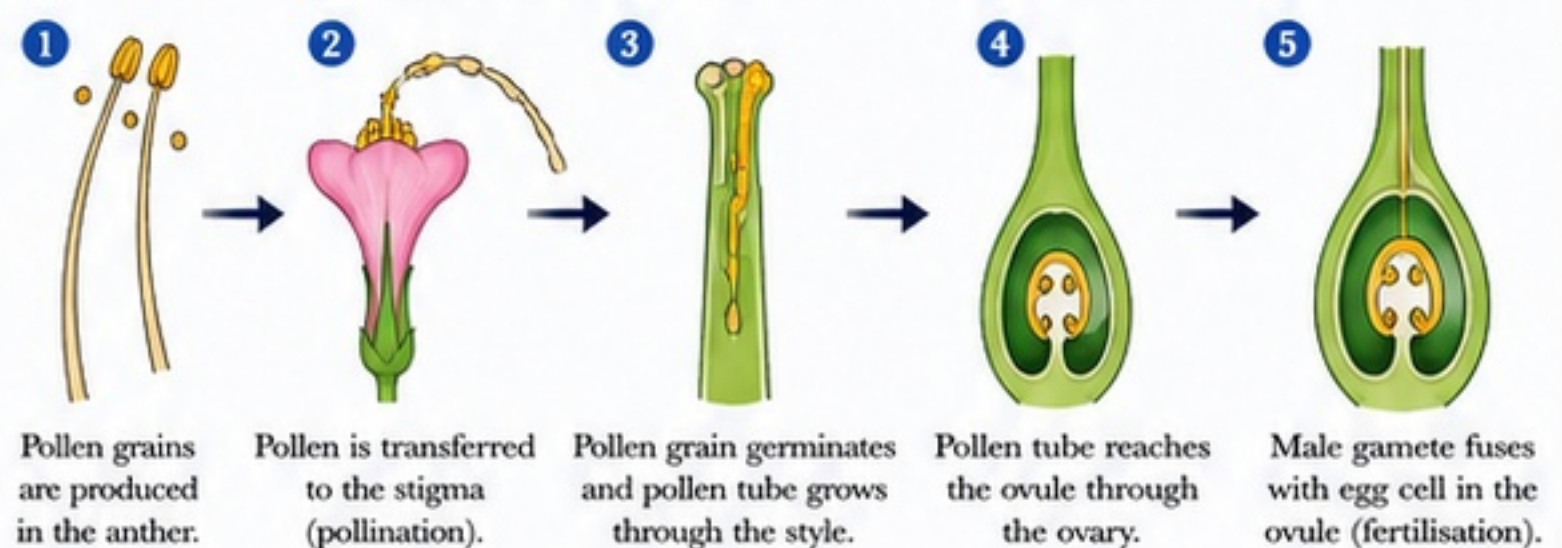
10. From ovary to fruit

- After fertilisation, the ovary develops into a fruit.
- Fruit protects seeds and helps in their dispersal.

STRUCTURE OF A FLOWER



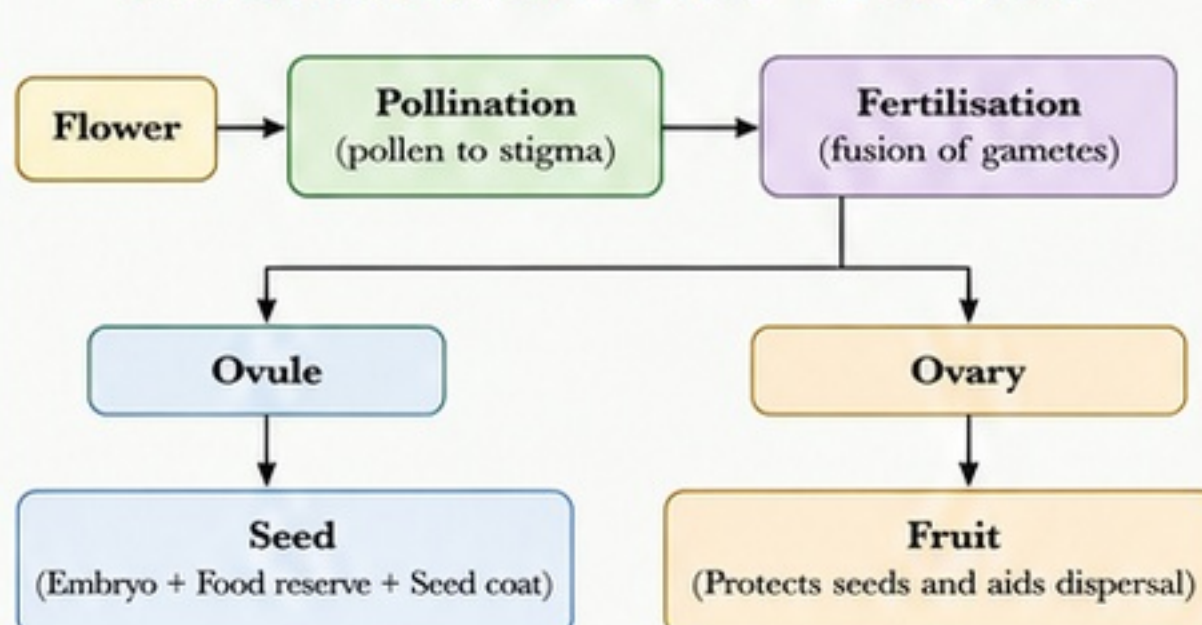
POLLINATION AND FERTILISATION - SEQUENCE



SELF-POLLINATION vs CROSS-POLLINATION

Basis	Self-Pollination	Cross-Pollination
Definition	Pollen transfer within the same flower or same plant.	Pollen transfer between flowers of different plants of the same species.
Genetic makeup of offspring	Little variation; offspring are similar to parents.	More variation; offspring are different from parents.
Dependence on agents	Usually does not require external agents.	Mostly requires agents (wind, water, animals).
Examples	Pea, wheat, rice, tomato	Maize, papaya, mango, sunflower
Advantages	Assured seed set; maintains pure lines.	Promotes variation; helps in adaptation and evolution.
Disadvantages	Less variation; may lead to inbreeding depression.	Uncertain seed set; depends on agents.

FLOWER TO SEED AND FRUIT - FLOWCHART

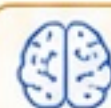


KEY IDEA

Sexual reproduction in flowering plants involves pollination and fertilisation, leading to the formation of seeds (from ovules) and fruits (from ovary). It ensures continuity of species and variation.

EXAM TIPS

- Always label parts of a flower neatly.
- In diagrams, show pollen tube, ovule, embryo sac, and ovary clearly.
- Write differences between self and cross pollination in a table.
- Flowcharts of processes are high-scoring in board exams.



QUICK RECALL

- Male gamete – Pollen grain
- Female gamete – Egg cell
- Pollination → Fertilisation → Seed & Fruit
- Ovule → Seed
- Ovary → Fruit
- Self-pollination – same plant; Cross-pollination – different plant
- Agents – Wind, Water, Animals (insects, birds, bats)

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SECTION E – GERMINATION, SEED FORMATION AND REPRODUCTIVE STRATEGIES IN PLANTS

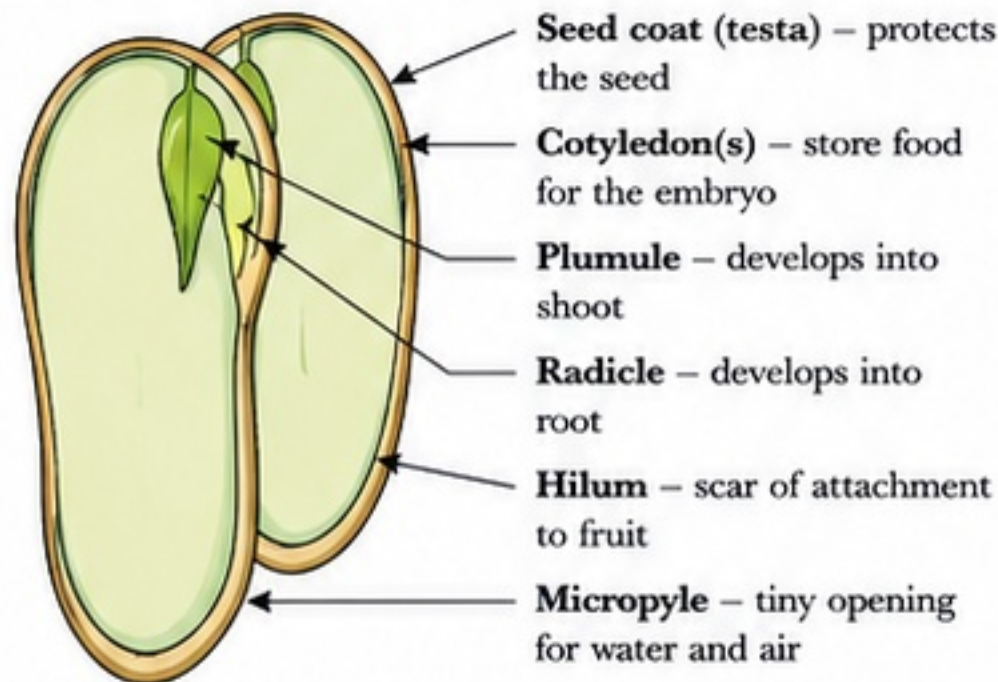
1. Post-fertilisation changes in a flower

- After fertilisation, ovules and ovary undergo changes to form seeds and fruits.
- Petals, sepals, stamens and style usually wither and fall off.
- **Ovary enlarges to form fruit.**
- **Ovules develop into seeds.**

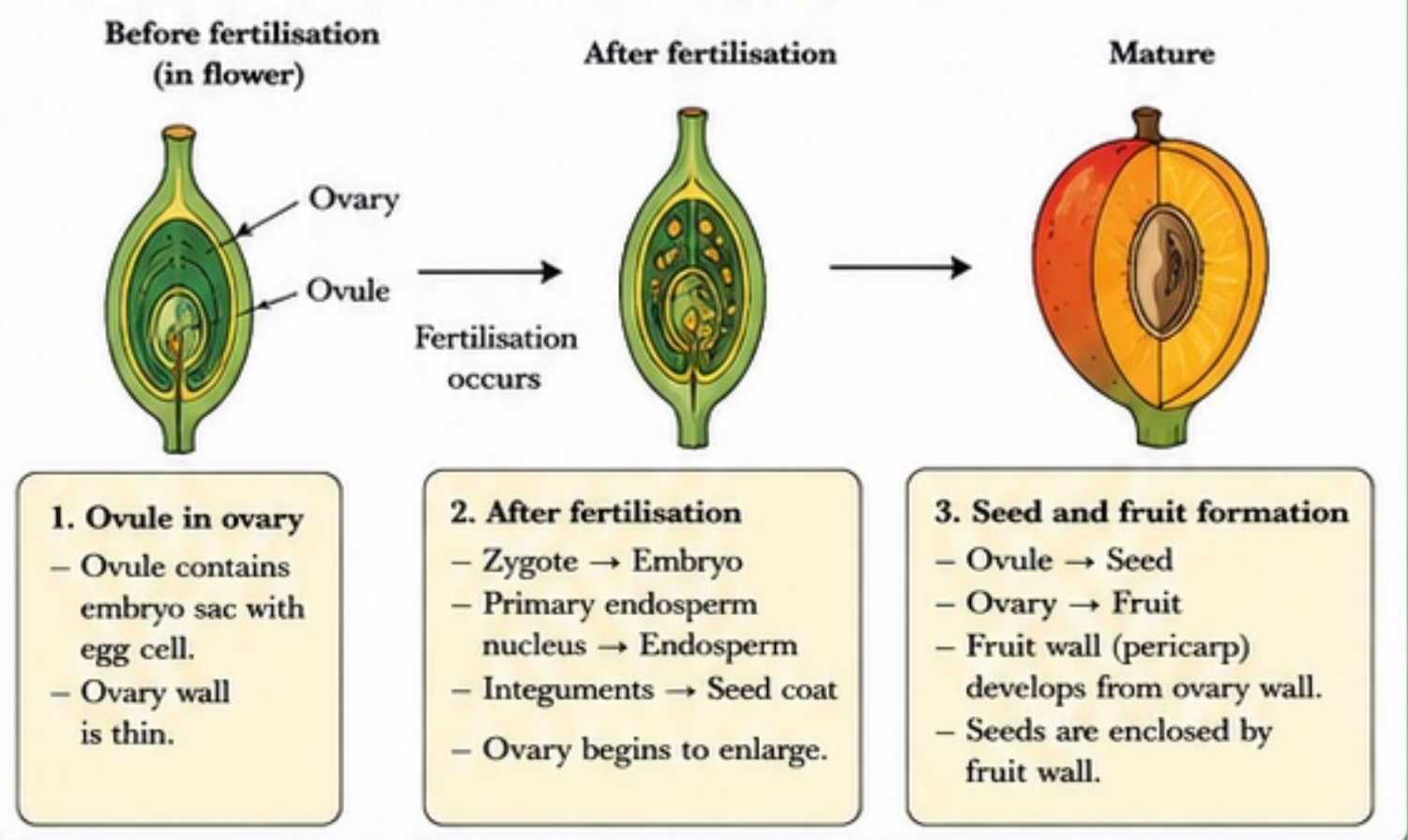
2. Development of seed and fruit

- Each ovule contains an embryo sac.
- Fertilised egg (zygote) develops into embryo.
- Primary endosperm nucleus develops into endosperm (food reserve).
- Integuments of ovule form seed coat.
- **Ovule → Seed → Ovary → Fruit**

3. Structure of a typical seed (e.g., bean)

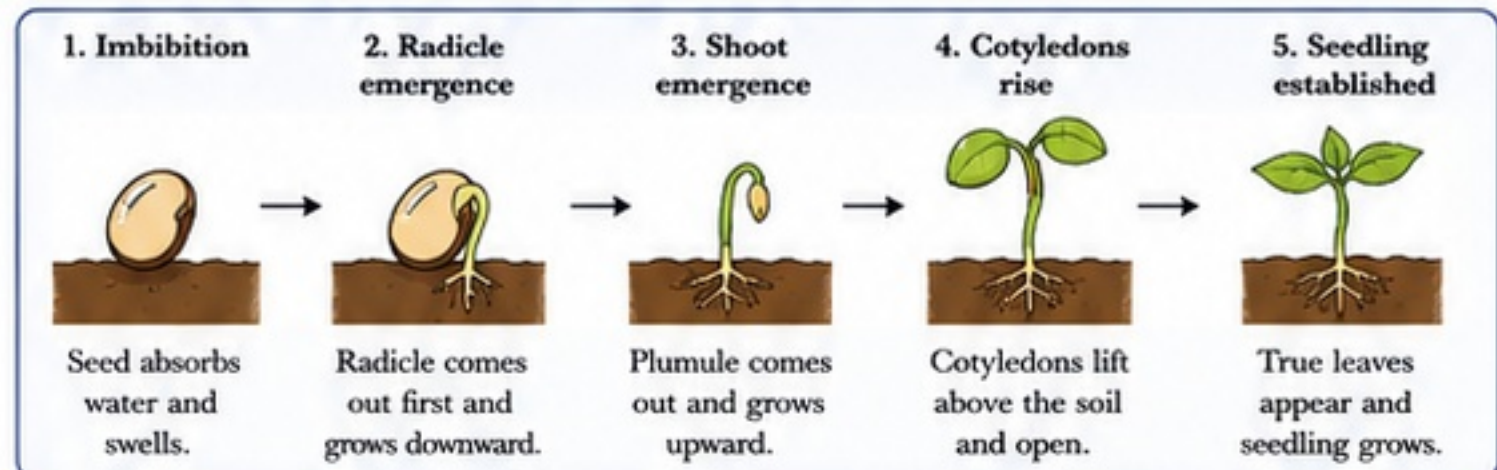


4. From Ovule to Seed and Ovary to Fruit



5. Germination of seeds

- **Germination** is the process by which a seed grows into a seedling.
- **Conditions necessary for germination:**
 1. **Water** (moisture) – softens seed coat and activates enzymes.
 2. **Oxygen** (air) – needed for respiration to release energy.
 3. **Suitable temperature** – each plant has an optimum range.
- **Sequence of germination (e.g., dicot seed)**



6. Seed dispersal – Methods and Examples

Method	How it works	Examples of Plants	Adaptations / Features
Wind dispersal (Anemochory)	Light seeds/fruits are carried away by wind.	Dandelion, Cotton, Maple, Drumstick (Moringa)	Very light; winged or feathery structures; hairy (pappus).
Water dispersal (Hydrochory)	Fruits/seeds float on water and are carried to new places.	Coconut, Lotus, Water lily	Spongy/hollow outer coat; fibrous husk; waterproof.
Animal dispersal (Zoochory)	Seeds/fruits are eaten and excreted elsewhere or get stuck to body fur.	Guava, Berry fruits, Xanthium (cocklebur), Banyan	Fleshy, tasty fruits; hard seeds pass through digestive system; hooks or sticky surfaces.
Explosion (Ballochory)	Dry fruits burst open suddenly and scatter seeds.	Pea, Balsam, Castor, Bean	Dry, elastic fruit wall builds tension; splits open on ripening.

7. Significance of seed dispersal

- Prevents overcrowding near parent plant.
- Seeds reach new places with good resources.
- Reduces competition for light, water, nutrients and space.
- Ensures survival and spread of the species.
- Helps in colonisation of new habitats.

KEY IDEA

Seeds are the survival capsules of plants! They remain dormant under unfavourable conditions and germinate when conditions become favourable. Dispersal ensures seeds find safe places to grow.

8. Why does sexual reproduction create variation in plants?

- In sexual reproduction, new individuals form by the fusion of gametes from two parents.
- During gamete formation, meiosis causes recombination (crossing over and independent assortment), creating new gene combinations.
- Random fertilisation (any pollen can fertilise any ovule) further mixes genetic material.
- Result: Offspring inherit a unique combination of traits → variation.

EXAM TIPS

- For diagrams, label clearly: ovule parts, seed parts, and germination stages.
- Remember: Water + Oxygen + Suitable Temperature = Germination.
- In seed dispersal answers, write method + mechanism + examples.
- Distinguish ovule vs seed and ovary vs fruit in 2–3 lines.
- Use flowcharts for processes: fertilisation → seed/fruit, germination.



QUICK RECALL

- After fertilisation: Ovule → Seed; Ovary → Fruit; other parts of flower fall off.
- Seed dispersal methods: Wind, Water, Animals, Explosion.
- Sexual reproduction produces variation due to recombination and random fertilisation.
- Germination needs: Water, Oxygen, Suitable temperature.

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SECTION F - ADOLESCENCE, PUBERTY AND MALE REPRODUCTIVE SYSTEM

1. Adolescence

- The period between childhood and adulthood.
- Body undergoes physical, mental and emotional changes.
- Prepares the body for reproduction and adult responsibilities.

2. Puberty

- The stage when the body starts maturing and becomes capable of reproduction.
- Triggered by hormones released by endocrine glands.

3. Secondary sexual characters in males

- Growth of hair in armpits, chest, face and pubic region.
- Voice deepens due to enlargement of larynx.
- Broadening of shoulders and increased muscle mass.
- Increased activity of sweat and sebaceous glands.
- Growth spurt (rapid increase in height).
- Development of Adam's apple.

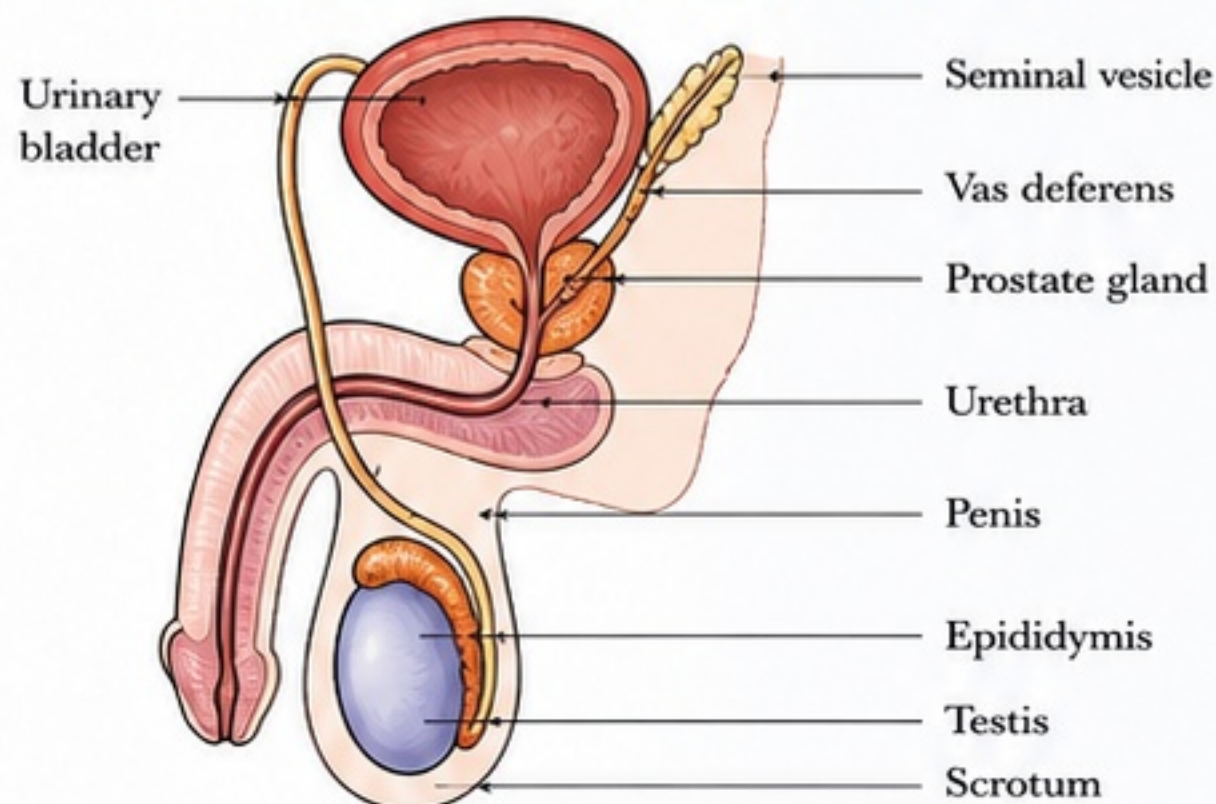
4. Role of hormones

- **Testosterone** (from testes) is the main hormone in males.
- It brings about changes of puberty, development of secondary sexual characters and production of sperms.
- Hormones act as chemical messengers in the body.

5. Onset of reproductive maturity

- In boys: usually between 12 – 16 years.
- By this time, the testes start producing sperms and the male becomes capable of reproduction.

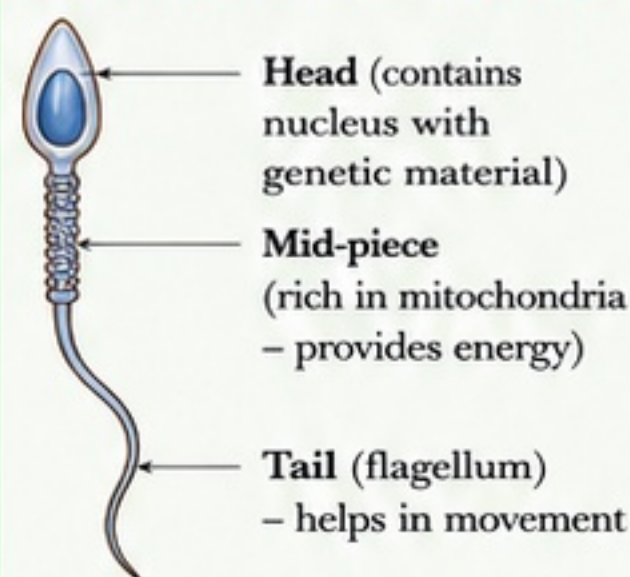
DIAGRAM: MALE REPRODUCTIVE SYSTEM



The male reproductive system consists of external organs (scrotum, penis) and internal organs (testes, ducts and glands).

Organ	Function
1. Testes	• Pair of oval organs present in the scrotum. • Produce sperms and the hormone testosterone.
2. Scrotum	• Sac of skin outside the abdomen that holds the testes. • Keeps testes at a temperature 2–2.5°C lower than body temperature, which is ideal for sperm production.
3. Epididymis	• Coiled tube on the surface of each testis. • Stores sperms and helps in their maturation.
4. Vas deferens (Sperm duct)	• Thick muscular tube. • Transports mature sperms from epididymis to urethra.
5. Seminal vesicles	• Pair of glands that secrete a fluid rich in fructose. • Provides energy to sperms and makes up a major part of semen.
6. Prostate gland	• Gland that adds a milky alkaline fluid to semen. • Helps in neutralising acidity in urethra and activates sperms.
7. Urethra	• Common passage for urine and semen. • Carries semen to the outside during ejaculation.
8. Penis	• External organ used to deliver semen into the female reproductive tract. • Made of spongy tissue; becomes erect during sexual arousal.

DIAGRAM: SPERM STRUCTURE



Sperm is a male gamete. It is microscopic, motile and has a streamlined structure to swim towards the egg.



Seminiferous tubules

9. SPERM (SHUKRAANU) PRODUCTION

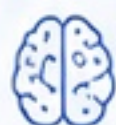
- Begins at puberty and continues throughout life.
- Occurs in the seminiferous tubules of testes.
- Process called spermatogenesis.
- Takes about 64–72 days for formation and maturation of sperms.
- Millions of sperms are produced daily.
- Sperms are released along with seminal fluid forming semen during ejaculation.

KEY IDEA

Testosterone brings about changes of puberty, growth of secondary sexual characters and production of sperms. Sperms are produced continuously in testes and are essential for reproduction.

EXAM TIPS

- Learn functions of each part in the male reproductive system.
- Label diagrams neatly and write short functions.
- Use flow arrows when describing the path of sperm.
- Write differences between primary and secondary sexual characters.



QUICK RECALL

- Adolescence – transition to adulthood.
- Puberty – onset of reproductive maturity.
- Testosterone – main male hormone.
- Spermatogenesis – formation of sperms in testes.
- Path of sperm – Testes → Epididymis → Vas deferens → Urethra → Penis.



REPRODUCTIVE HEALTH & HYGIENE

- Maintain personal hygiene; keep genital area clean and dry.
- Wear clean, well-fitted undergarments.
- Eat nutritious food, exercise and get enough sleep.
- Avoid tobacco, alcohol and harmful substances.
- Discuss concerns with parents/teachers or a trusted adult.
- Respect yourself and others; practise safe and responsible behaviour.

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SECTION G - FEMALE REPRODUCTIVE SYSTEM, MENSTRUATION AND FERTILISATION

1. Female reproductive system

It is the system in females that produces eggs (ova), receives sperm, supports fertilisation and nourishes the developing baby.

Main parts:

- **Ovaries** – produce ova and hormones (estrogen, progesterone).
- **Fallopian tubes (oviducts)** – carry ovum from ovary to uterus; site of fertilisation.
- **Uterus** – accepts the embryo and supports its development.
- **Cervix** – lower narrow part of uterus that opens into vagina.
- **Vagina** – muscular canal that receives penis/sperms and acts as birth canal.

2. Ovum formation (Oogenesis)

- At birth, ovaries contain thousands of immature eggs.
- From puberty, about one egg matures each month.
- One mature ovum is released from an ovary (ovulation) into the fallopian tube.

3. Menstrual cycle (about 28 days)

- **Day 1–5: Menstruation** – the lining of uterus breaks down and comes out as blood and tissue through vagina.
- **Day 6–13: Follicular phase** – egg matures; uterine lining rebuilds.
- **Day 14: Ovulation** – mature ovum released.
- **Day 15–28: Luteal phase** – lining becomes thick and spongy, ready for implantation. If no fertilisation, lining breaks down and next cycle begins.

4. Fertilisation

- When a sperm fuses with ovum, it is called fertilisation.
- It usually occurs in the fallopian tube (oviduct).
- Requires presence of both gametes; usually within 24 hours after ovulation.

5. Zygote formation and early development

- After fertilisation, a single cell called **zygote** is formed.
- Zygote divides repeatedly while moving towards the uterus: 2 cells → 4 cells → 8 cells → morula → blastocyst.

6. Implantation

- Blastocyst reaches uterus (about 5–6 days after fertilisation) and gets implanted in the thick uterine lining (endometrium).
- Implantation usually occurs on the upper inner wall of the uterus.

7. Embryo, placenta and foetus

- After implantation, embryo continues to grow.
- **Placenta develops:** a special disc-like structure that connects embryo with uterine wall.
- **Foetus:** After about 8 weeks, the growing baby is called a foetus. Major organs develop.
- **Umbilical cord** connects foetus to placenta.

8. Gestation

- The period from implantation to birth is called gestation.
- In humans, gestation period is about 38–40 weeks (about 9 months). Baby is then born.

DIAGRAM: FEMALE REPRODUCTIVE SYSTEM

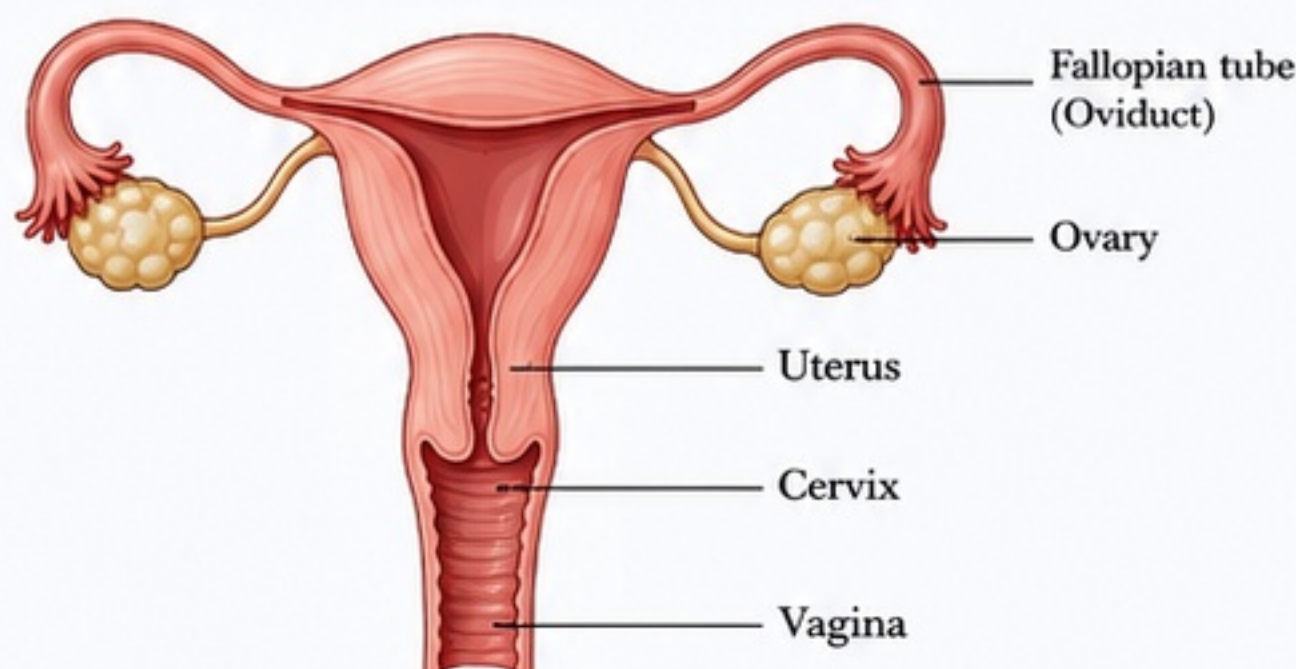
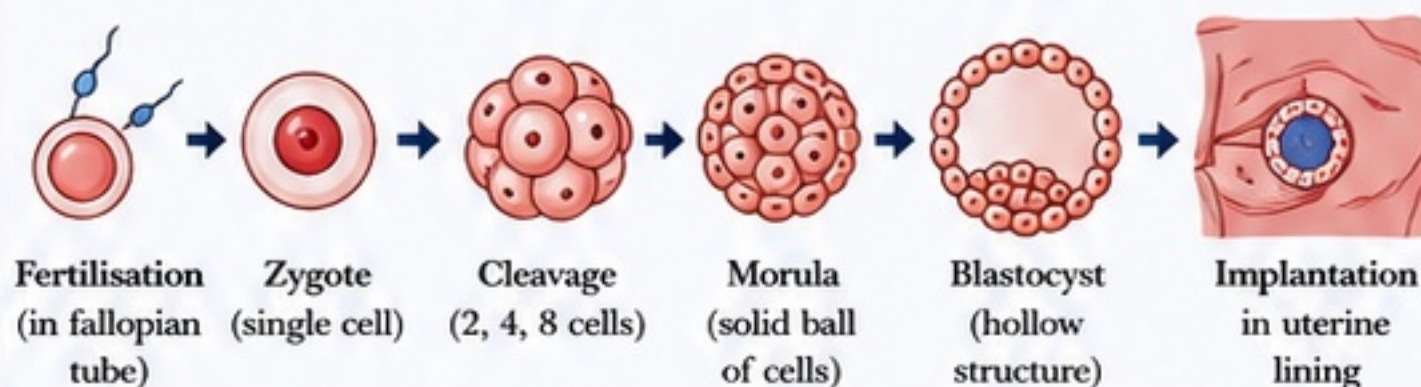


TABLE: PARTS OF FEMALE REPRODUCTIVE SYSTEM AND THEIR FUNCTIONS

Part	Function
Ovaries	Produce ova (eggs) and hormones like estrogen and progesterone.
Fallopian tubes (Oviducts)	Pick up ovum from ovary; transport it to uterus. Site of fertilisation.
Uterus	Thick muscular organ; site of implantation and development of embryo/foetus.
Cervix	Narrow opening of uterus into vagina; keeps uterus closed during pregnancy.
Vagina	Receives penis/sperms during intercourse; acts as birth canal during delivery.

FLOW DIAGRAM: FERTILISATION TO IMPLANTATION



IMPORTANT BOARD POINTS

- **Fertilisation** occurs in the fallopian tube (oviduct), usually in the ampulla.
- **Implantation** occurs in the thick uterine lining (endometrium).
- **Placenta** acts as a link between mother and foetus. It:
 - Provides oxygen and nutrients to the foetus.
 - Removes carbon dioxide and other wastes from foetus.
 - Produces hormones that maintain pregnancy.



KEY IDEA

The female reproductive system produces eggs and hormones, supports fertilisation, nourishes the embryo through placenta, and brings new life through a full-term pregnancy.



EXAM TIPS

- Learn the diagram and label all parts correctly.
- Remember: Fertilisation – in fallopian tube; Implantation – in uterus.
- Write menstrual cycle in proper order with days.
- In answers, use flow/step format for fertilisation and development.

9. EVERYDAY EXAMPLES



Menstruation

- The monthly bleeding is natural; it shows the body is healthy.
- Use clean and hygienic methods during periods.



Care during pregnancy

- Eat nutritious food, rest well, exercise lightly.
- Avoid heavy work, stress and harmful habits.
- Regular check-ups help keep mother and baby healthy.



QUICK RECALL

- Q1. Where is fertilisation in humans? A. Fallopian tube (oviduct).
Q2. Where does implantation occur? A. Uterine lining (endometrium).
Q3. What is placenta? A. Link between mother and foetus.
Q4. Gestation period in humans? A. About 38–40 weeks.
Q5. What is the function of placenta? A. Provides nutrients & O₂, removes wastes, produces hormones.

Ch 7: How do Organisms Reproduce?

Part 1 of 1 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

SECTION H - REPRODUCTIVE HEALTH, CONTRACEPTION AND RESPONSIBLE PARENTHOOD

1. Need for family planning

- Population explosion puts pressure on resources like food, water, housing, education and health care.
- Leads to unemployment, poverty, malnutrition and environmental degradation.
- Family planning helps to achieve a balance between population and resources for a better quality of life.

2. Importance of small family norm

- Ensures proper care, nutrition, education and emotional support to children.
- Promotes health and well-being of parents, especially mothers.
- Helps in building a healthy and progressive society.

3. Reproductive health

- Complete physical, mental and social well-being in all matters related to the reproductive system.
- Includes awareness, proper care, nutrition, hygiene and access to health services.

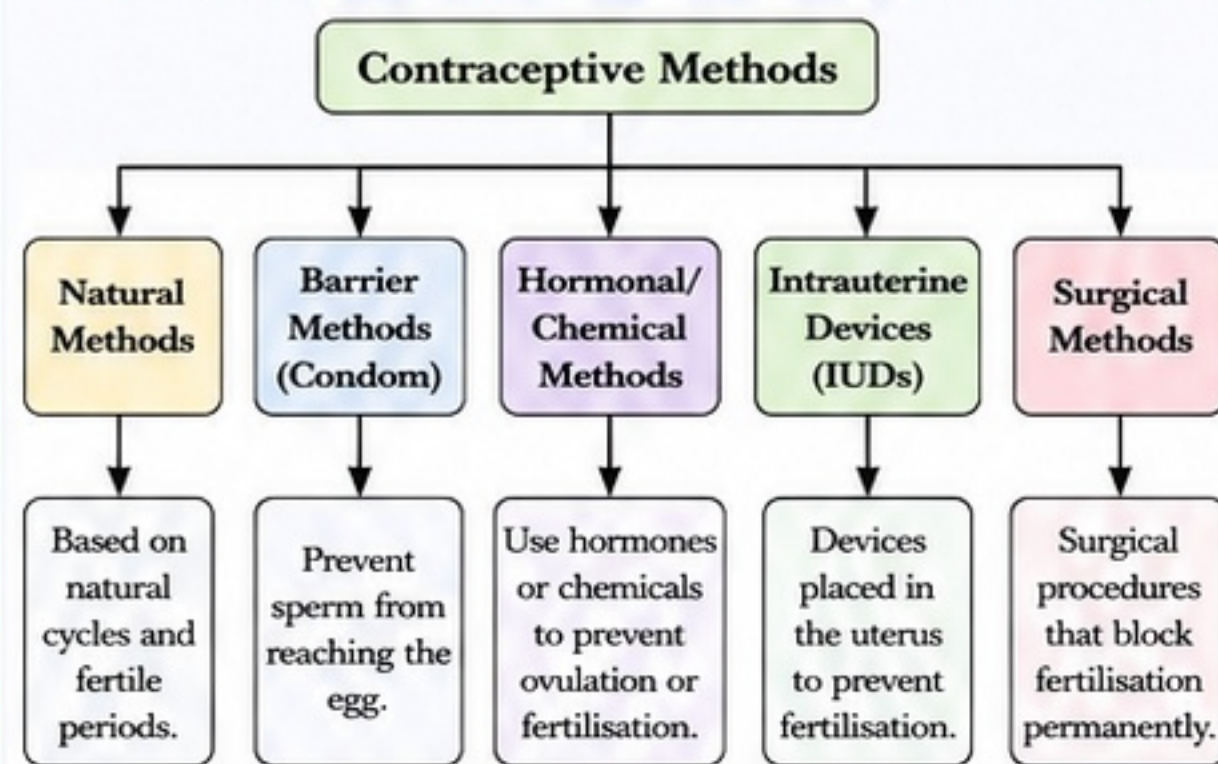
4. Safe motherhood

- Ensures health and safety of the mother before, during and after childbirth.
- Requires regular check-ups, balanced diet, rest, clean environment, skilled medical care and immunisation.
- Institutional delivery and postnatal care reduce risks and complications.

5. Contraception

- Contraception means preventing pregnancy.
- It helps couples plan their family and maintain the health of the mother.
- Various methods are available; choice depends on health, suitability, convenience and guidance from health professionals.

CLASSIFICATION OF CONTRACEPTIVE METHODS



Note: Only condoms (barrier method) help prevent sexually transmitted infections (STIs) as well as unwanted pregnancy.

SEXUALLY TRANSMITTED INFECTIONS (STIs)

- STIs are infections spread mainly through unprotected sexual contact.
- Common STIs:
 - Gonorrhoea** – Caused by bacteria (*Neisseria gonorrhoeae*); affects reproductive organs and can cause infertility if untreated.
 - Syphilis** – Caused by bacteria (*Treponema pallidum*); causes sores, fever and serious complications if untreated.
 - HIV/AIDS** – Caused by Human Immunodeficiency Virus; weakens the immune system and can lead to AIDS.
- Prevention:** Use of condoms, one faithful partner, avoidance of risky behaviour, blood screening and awareness.

COMPARISON OF CONTRACEPTIVE METHODS

Method	Examples	How it works	Advantages	Limitations
Natural Methods	Fertility awareness (safe period), withdrawal, lactational amenorrhoea	Avoids intercourse during fertile days or before ejaculation; relies on natural cycles	No cost; no side effects; promotes awareness	Less reliable; requires good knowledge and self-control
Barrier Methods (Condom)	Male condom, female condom	Prevents sperms from reaching the egg	Easy to use; no hormones; protects against STIs	Must be used correctly every time; may tear or slip
Hormonal/Chemical Methods	Oral pills, injectables, implants, contraceptive patches, spermicides	Hormones prevent ovulation or thicken cervical mucus; spermicides kill sperms	Highly effective; variety of options	Possible side effects (nausea, headache, weight changes); needs medical advice
Intrauterine Devices (IUDs)	Copper-T (CuT 380A), Hormonal IUD	Device in uterus prevents fertilisation or implantation	Long-term protection; highly effective; reversible	May cause cramps, bleeding initially; must be inserted by trained professional
Surgical Methods	Vasectomy (male), Tubectomy (female)	Cuts or seals vas deferens/fallopian tubes to block sperms or egg passage	Permanent; very effective; no daily/regular action needed	Irreversible (or difficult to reverse); requires minor surgery

MYTH vs FACT

Myth	Fact
Contraceptives cause permanent infertility.	Most methods are reversible. Fertility returns after stopping (except rare cases like surgery).
Condoms reduce pleasure.	Proper use of condoms does not reduce pleasure and ensures safety.
Only women should use contraception.	Contraception is a joint responsibility of both partners.
Taking pills once can prevent pregnancy permanently.	Pills must be taken regularly as advised; missing doses reduces effectiveness.



KEY IDEA

- Reproductive health and responsible parenthood lead to healthy individuals and a healthy society.
- Contraception helps in preventing unwanted pregnancy and planning the number and spacing of children.
- Condoms are the only method that also protects against STIs.



EXAM TIPS

- Learn the classification of contraceptive methods with examples.
- Remember: Condom = dual protection (pregnancy + STIs).
- Write advantages and limitations in tabular form.
- Use flowcharts for classification answers.
- Link answers to health, economy and quality of life wherever possible.



QUICK RECALL

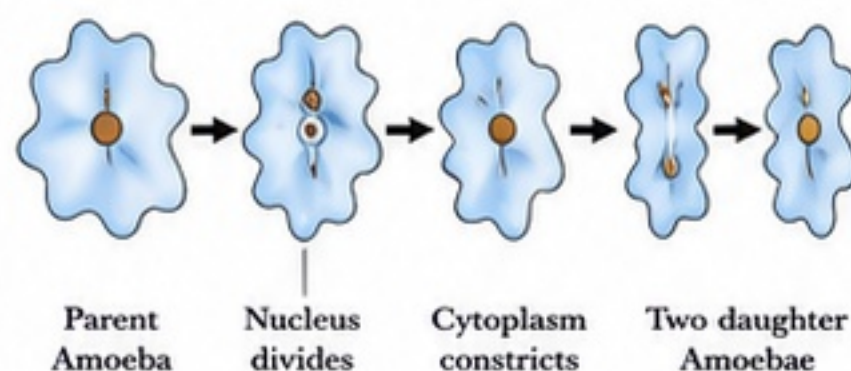
- Need for family planning – population control, resource balance, better life.
- Small family norm – better care, education, health and future.
- Reproductive health – physical, mental and social well-being.
- Safe motherhood – care before, during, after childbirth.
- Contraception – prevents pregnancy; many methods available.
- Condom – also protects against STIs (e.g., gonorrhoea, syphilis, HIV/AIDS).

Ch 7: How do Organisms Reproduce?

Part 1 of 1 | Complete Notes | Deep Concepts | Diagrams | Tables | Exam Tips | Centum Focus

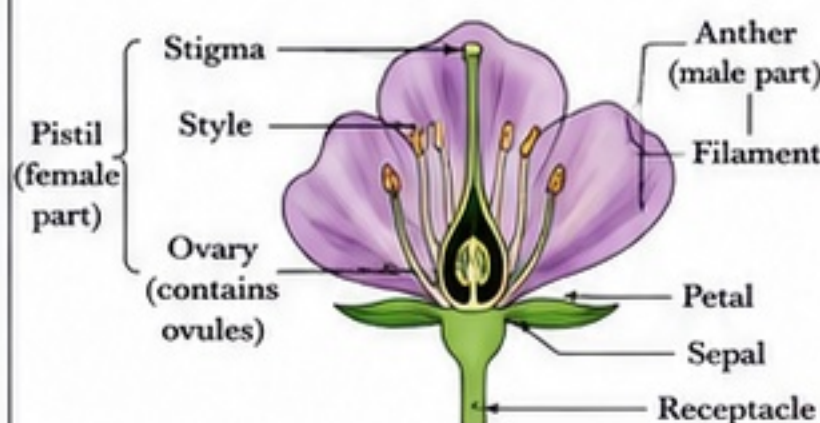
SECTION I – IMPORTANT DIAGRAMS, COMPARISONS AND ANSWER FRAMES

1. BINARY FISSION IN AMOEBA



- Amoeba reproduces by binary fission.
- The nucleus divides into two.
- The cytoplasm splits to form two daughter cells.
- Occurs in favourable conditions and ensures rapid increase in number.

2. FLOWER – REPRODUCTIVE PARTS AND POLLINATION



Pollination – Transfer of pollen grains from anther to stigma.

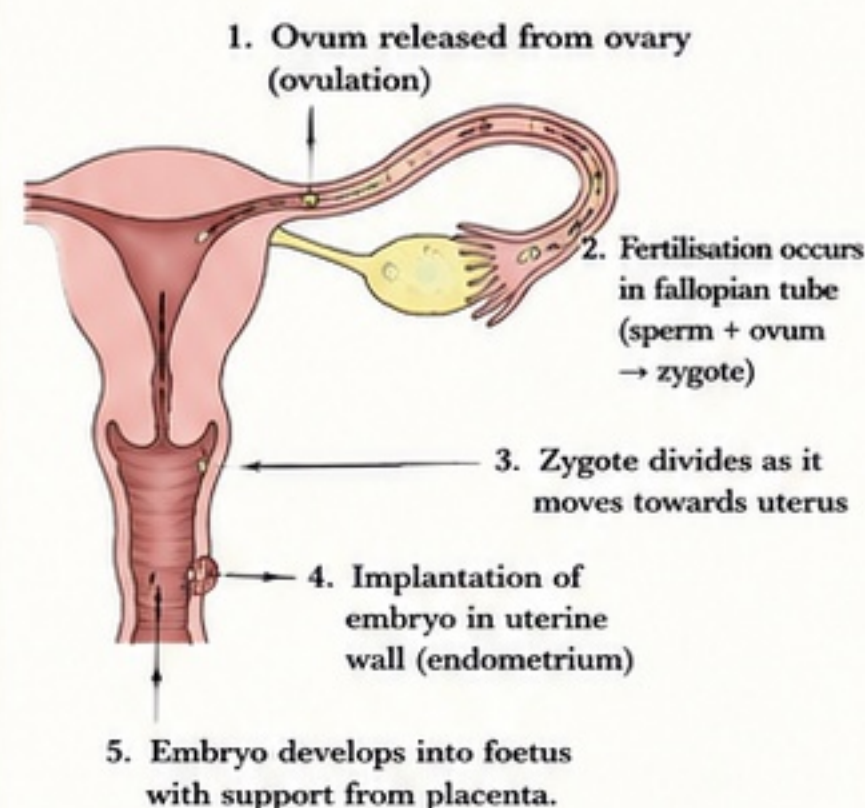
Self-pollination

From same flower or another flower on the same plant.

Cross-pollination

From a flower on one plant to a flower on another plant.

3. HUMAN FERTILISATION AND IMPLANTATION – OVERVIEW



ANSWER FRAMES

1. Define reproduction.

Reproduction is the biological process by which new organisms are produced from their parents of the same kind.

2. Differentiate between asexual and sexual reproduction.

Asexual reproduction involves one parent and no fusion of gametes; offspring are genetically identical. Sexual reproduction involves two parents and fusion of gametes; offspring show variation.

3. Explain vegetative propagation with an example.

Vegetative propagation is a type of asexual reproduction in plants where new plants grow from vegetative parts like root, stem or leaf. Example: Potato – new plant grows from the 'eyes' (buds) on the tuber.

4. Differentiate between pollination and fertilisation.

Pollination is the transfer of pollen from anther to stigma. Fertilisation is the fusion of male gamete from pollen with the female gamete in the ovule to form a zygote.

5. State the role of placenta during pregnancy.

Placenta connects the foetus to the uterine wall and allows transfer of nutrients and oxygen from the mother to the foetus and removal of carbon dioxide and wastes from the foetus.

TWO-MARK QUESTIONS (2M)

- Q: What is asexual reproduction? Give one example.
A: Reproduction involving one parent without fusion of gametes.
Example: Binary fission in Amoeba.
- Q: List two advantages of sexual reproduction.
A: (i) Offspring show variations. (ii) Helps in adaptation and survival in changing environments.
- Q: Name any two agents of pollination.
A: Wind, insects, water, animals (any two).
- Q: Where does fertilisation take place in human female?
A: In the fallopian tube (oviduct).

THREE-MARK QUESTIONS (3M)

- Q: Describe binary fission in Amoeba with a labeled diagram.
A: (i) Nucleus divides into two. (ii) Cytoplasm constricts in the middle. (iii) Two daughter Amoebae are formed. (iv) Diagram as shown above.
- Q: What is pollination? Explain its types with examples.
A: Transfer of pollen from anther to stigma is pollination.
Types: (i) Self-pollination – in same flower or same plant (e.g., pea). (iii) Cross-pollination – in different plants of same species (e.g., maize, papaya).
- Q: Explain the process of fertilisation and implantation in humans.
A: Fertilisation occurs in the fallopian tube when sperm fuses with ovum to form zygote. The zygote divides and moves to the uterus. It implants in the uterine wall (endometrium) with the help of placenta.

COMPARISON TABLES

(A) ASEQUAL VS SEXUAL REPRODUCTION

Feature	Asexual	Sexual
Parents involved	One	Two
Gametes	Not involved	Involved
Fusion of gametes	Absent	Present
Offspring	Genetically identical (clones)	Genetically varied
Rate	Usually fast	Usually slow
Variation	Very little	High
Example	Binary fission, Budding, Tuber	Human beings, Flowering plants

(B) POLLINATION VS FERTILISATION

Feature	Pollination	Fertilisation
Meaning	Transfer of pollen from anther to stigma	Fusion of male and female gametes
Site	Stigma of flower	Ovule (inside ovary)
Agents	Wind, insects, water, animals	Internal process/ no external agent
Result	No zygote formed	Zygote is formed
Necessity	Precedes fertilisation	Occurs after pollination

(C) MALE VS FEMALE REPRODUCTIVE SYSTEM (HUMAN)

Feature	Male	Female
Main organs	Testes, Vas deferens, Penis	Ovaries, Fallopian tubes, Uterus, Vagina
Gamete	Sperm	Ovum
Gamete formation	Continuous after puberty	Cyclic (one ovum released each month)
Site of gamete formation	Testes	Ovaries
Functions	Produces sperm and male hormones	Produces ova, female hormones; site of fertilisation & foetal development



Reproduction • Offspring • Continuity • Asexual • Sexual • Gamete • Fertilisation • Pollination • Self-pollination
Cross-pollination • Binary fission • Budding • Vegetative propagation • Ovule • Zygote • Embryo • Foetus • Implantation
Placenta • Variation • Adaptation • Survival

SECTION J - MASTER REVISION SHEET, NCERT TRAPS AND CENTUM FOCUS

1. COMPACT DEFINITIONS

Term	Definition
Reproduction	Biological process by which organisms produce new individuals of their own kind.
Asexual reproduction	Reproduction involving a single parent; no fusion of gametes; offspring are genetically identical (clones).
Sexual reproduction	Reproduction involving two parents (or two gametes); fusion of gametes forms a zygote; offspring show variation.
Pollination	Transfer of pollen grains from the anther to the stigma of a flower.
Fertilisation	Fusion of male gamete (sperm) with female gamete (ovum) to form a zygote.
Vegetative propagation	A type of asexual reproduction in plants using vegetative parts like root, stem or leaf.
Puberty	Stage in human adolescence when reproductive organs mature and secondary sexual characters appear.
Menstruation	Monthly shedding of the inner lining of the uterus through the vagina if fertilisation does not occur.
Contraception	Methods or devices used to prevent pregnancy.
Placenta	A temporary disc-like structure in the uterus that connects the foetus to the mother and allows exchange of nutrients, gases and wastes.

4. NCERT TRAPS & MISCONCEPTIONS

Trap / Misconception	Correct Explanation
Asexual reproduction occurs only in animals.	False. It is very common in plants (e.g., potato, Bryophyllum).
Budding and binary fission are the same.	Budding is an outgrowth (e.g., Hydra); binary fission is splitting into two equal parts (e.g., Amoeba).
Pollination and fertilisation are the same.	Pollination is transfer of pollen; fertilisation is fusion of gametes.
Fertilisation occurs in the stigma of the flower.	False. It occurs in the ovule inside the ovary.
Humans can reproduce asexually.	False. Humans reproduce only by sexual reproduction.
Zygote develops into baby in the fallopian tube.	False. Zygote forms in the fallopian tube but develops in the uterus.
Menstruation occurs due to fertilisation.	False. It occurs when fertilisation does not take place.
All pollination is done by insects.	False. It can be by wind, water, birds, bats, and sometimes by self.

6. SUPER QUICK RECALL – 10 MUST REMEMBER

- 1 Reproduction ensures continuity of life.
- 2 Asexual reproduction – one parent, no gamete fusion; offspring are clones.
- 3 Sexual reproduction – two parents/two gametes; offspring show variation.
- 4 Pollination is transfer of pollen; fertilisation is fusion of gametes.
- 5 Self-pollination – same flower/same plant; cross-pollination – different plants.
- 6 In humans, fertilisation occurs in fallopian tube; development in uterus.
- 7 Puberty is the stage of sexual maturation.
- 8 Menstruation occurs when fertilisation does not take place.
- 9 Contraception prevents pregnancy and helps in family planning.
- 10 Variation helps species adapt and survive in changing environments.

2. MASTER SUMMARY – THE BIG PICTURE

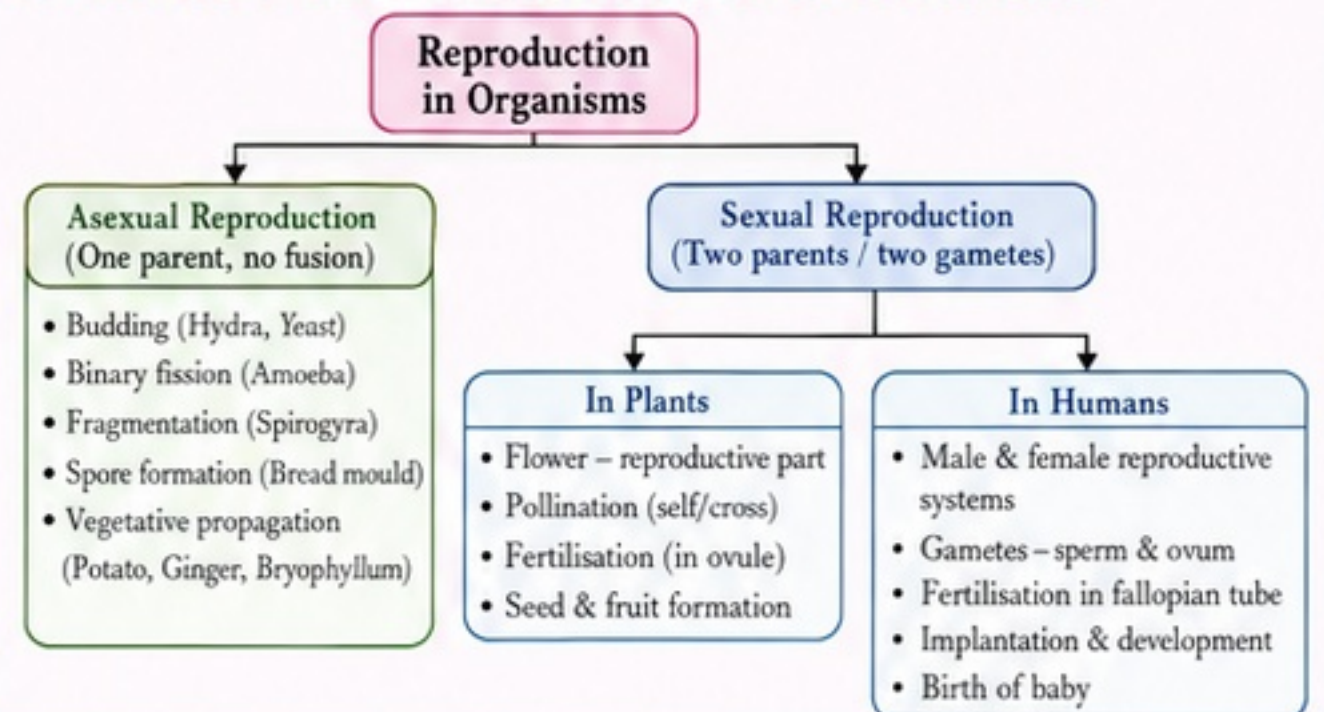
- Reproduction ensures continuity of life and survival of species.
- Two main types: asexual (one parent, no gamete fusion) and sexual (two parents/two gametes, fusion occurs).
- Asexual methods in plants: budding, binary fission, fragmentation, spore formation, vegetative propagation.
- Sexual reproduction in plants: flowers are reproductive organs; pollination → fertilisation → seed/fruit formation.
- Pollination may be self (same flower/plant) or cross (between different plants).
- Sexual reproduction in humans involves male and female reproductive systems, gametes, fertilisation in fallopian tube, implantation and foetal development.
- Adolescence brings puberty and changes in body. Menstruation is part of the reproductive cycle.
- Contraception helps in family planning and healthy living.
- Offspring produced by sexual reproduction show variation, helping species adapt and evolve.

3. COMPARE AND CONTRAST

Feature	Asexual Reproduction	Sexual Reproduction
Parents	One	Two (or two gametes)
Gametes	No gametes; no fusion	Gametes formed; fusion occurs
Offspring	Genetically identical (clones)	Genetically different (variation)
Speed	Generally fast	Generally slow
Energy required	Less	More
Examples	Budding, binary fission, vegetative propagation	Flowering plants, humans, most animals

Feature	Self-Pollination	Cross-Pollination
Definition	Pollen transfer within the same flower or same plant	Pollen transfer between flowers of different plants of the same species
Agents	Usually wind or self	Wind, water, insects, birds, etc.
Genetic variation	Low	High
Offspring	Less variation	More variation
Example	Pea, Wheat, Rice	Maize, Papaya, Mustard

5. CONCEPT MAP – REPRODUCTION IN ORGANISMS



7. CENTUM FOCUS – REVISE THIS BEFORE EXAM

- ☐ Types of reproduction – asexual & sexual (with examples).
- ☐ Asexual methods in plants – 5 methods with examples.
- ☐ Parts of a flower and functions.
- ☐ Steps of sexual reproduction in flowering plants (sequence).
- ☐ Difference between self and cross pollination (agents, variation).
- ☐ Human male and female reproductive systems (labels & functions).
- ☐ Path of sperm and ovum; site of fertilisation.
- ☐ Menstrual cycle – key points.
- ☐ Contraceptive methods – at least 4 examples.
- ☐ Key terms: zygote, gamete, ovule, placenta, embryo, spore.
- ☐ NCERT diagrams: flower (L.S.), human reproductive system, sperm, ovum, menstrual cycle.



Revise smart, label diagrams neatly, and score full marks.

